DIVIDEND POLICY DECISIONS

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STUDY GUIDE

NOTE:

- The contents of this material may be used for study by:
  - final year undergraduate students
  - Post-Graduate Diploma Students
  - MBF, MDF, MBA & M.Sc. students
- However, the following topic extensions are exclusively for:
  - M.Sc. Students:
    - Multi-stage dividend model,
    - Mathematical Proof of M&M Dividend Irrelevance Views
    - Students are also expected to explore framework of each equation
  - MBA, MBF & MDF Students:
    - Case Study
    - Survey of dividend payments of companies listed in NSE

LECTURE OBJECTIVES

- To enable students understand the meaning of dividend, dividend policy and other ways companies return cash to shareholders

- To enable students know whether dividend policy decision is a value maximizing exercise
  - Appreciate when investors are likely to prefer cash dividend to capital gains, and vice versa
  - Appreciate the views of different schools of thought regarding policy and value

- To evaluate real life dividend policy case to internalize the above theories, models and concepts

INTRODUCTION:
Definitions, Concepts

- Dividend in a general sense refers to payment made to shareholders from earning and capital as a reward for investment made in the shares of the company
  - Some authors have defined it within the narrow context of cash payment from earning made by a company
  - Cash distribution out of capital may be subject to strict regulation from regulatory authorities

- Dividend Policy is a formal guide of a firm's corporate behaviour in determining the type, amount and timing of dividend payment to shareholders

KEY POINTS, DATES & PROCEDURES TO NOTE

- Dividend declaration is down to decision of the Board while shareholders approve
- Decision to pay dividend is in practice simultaneously a dividend and financing decision
- Declaration Date: The date directors pass a resolution proposing a dividend
- Date of Record: The date in which all shareholders whose names appear in the books of the company are entitled to dividend

KEY POINTS, DATES & PROCEDURES TO NOTE

- Ex Div. Date: A date at or after which any buyer of shares will not be entitled to dividend. Date of Record is normally set a few days after this date to enable the Registrar update the books. During the period, the books may be closed

- Payment Date: This is the date in which dividend warrants are prepared and dispatched to shareholders of Record

- In an efficient market, no one is expected to abnormally profit from price movements purely as a result of dividend declaration and payment

TYPES OF DIVIDEND

- CASH DIVIDEND
  - Regular cash dividend
  - Special & extra cash dividend
  - Liquidating cash dividend

TYPES OF DIVIDEND

- STOCK DIVIDEND
  - Scrip or Bonus dividend: In this case, instead of cash, additional shares are ‘paid’ to shareholders on the basis of existing holdings. This may be used to ‘appease’ shareholders when cash can not be paid. Note however that where scrip are issued;
    - Total number of shares would increase
    - Book value of the firm remains unchanged
    - Market value of the firm remains unchanged
    - Nominal price per share remains unchanged
    - But market price per share is adjusted downwards

TYPES OF DIVIDEND

- Stock Split: This a situation where existing shares are split such that a shareholder receives increased quantity of shares of lower unit value in exchange. When a stock is split;
  - Total number of shares increases
  - Book value of the firm remains unchanged
  - Market value of the firm remains unchanged
  - Market price & nominal value per share are reduced proportionately

OTHER TYPES OF DIVIDEND

- Reverse Split: This occurs when “Consolidation” of shares takes place to reduce the number of shares outstanding and increase per-unit value
- Why firms may split shares or do reverse split (logical and illogical)
  - When market price is considered too expensive or low
  - Regulatory requirement for public listing in some jurisdictions
  - Merger & acquisition negotiations
  - To reduce cost of printing in the case of reverse splits
- Note that like cash dividends, bonuses and splits, on their own, do not add value to the firm in well functioning markets

OTHER TYPES OF DIVIDENDS

- **SHARE REPURCHASES**: Though not dividend, in the conventional meaning, it is another way firms may choose to return money to shareholders. It involves a buy back of shares held by shareholders for cash.
- Effects of corporate action:
  - Reduction in number of shares outstanding
  - Reduction in asset of cash in the company
  - Reduction in equity capital
- Why firms may repurchase shares:
  - When the firm has excess cash relative to positive NPV opportunities
  - When capital gains tax advantage exist for shareholders
  - When there exists restrictions on dividend payment
  - When the firm targets a certain debt/equity ratio
  - Though hardly a valid economic justification, a firm may seek to repurchase shares to manage share glut and improve per share performance data.

ILLUSTRATION 1

- Equity Nig. Ltd has the following capital structure before declaring stock dividend of 1 for every 5 previously held in 2020.
  - Paid up ordinary shares of N1.00 per share N100m
  - 5% debenture 2028 20m
  - Reserves 50m

    170m

  The shares were selling at N1.20 per share in the market.

  Required:

  a) what is the post stock dividend market price?

  b) How would the capital structure look like after stock dividend payment?

ILLUSTRATION

Solutions on scrip issue

- Note: In calculating Ex-Stock Div. market price, recall that market capitalization before and after stock dividend are equal-

  $N_b\ P_{mb}\ = N_a\ P_{ma}$ (eqn. 1)

  where N_b is no of shares before stock div = 100m

  N_a is no of shares after stock dividend = 100m+100m(1/5) = 120m

  P_{mb} is market price before stock dividend = N1.20 per share

  P_{ma} is market price after stock dividend

  $100\ (1.2) = 120\ (P_{ma})$

  P_{ma} = N120/120 shares

  = N1.00 per share

- New Capital Structure
  - Paid up ordinary shares of N1.00 per share 120m
  - 5% debenture 2028 20m
  - Reserves (50m – 20m) 30m

    170m

Observe that, though number of shares increased, total book equity remains same at N150,000 just as total capital remains N170,000

ILLUSTRATION 2

- Equity Nig. Plc has the following capital structure before undertaking a stock split of 2:1 in 2020:
  - Paid up ordinary shares of N1.00 per share N100m
  - 5% debenture 2028 20m
  - Reserves 50m

    170m

  The shares were selling at N1.20 per share in the market.
  Required:

  a) Determine market price per share
  b) Determine nominal price per share
  c) Determine the new capital structure

ILLUSTRATION

Solution on stock split

- Market price after stock split (P_{ma})

  $N_b\ P_{mb}$ = $N_a\ P_{ma}$

  1 x 1.2 = 2 P_{ma}

  P_{ma} = N0.60k per share

- Nominal Price after stock split (P_{na})

  $N_b\ P_{nb}$ = $N_a\ P_{na}$

  1 x 1.0 = 2 P_{na}

  P_{na} = N0.50k per share

- New Capital structure after stock split

– 200m Ordinary shares of N0.50 per share	N100m
– 5% debenture 2028	20m
– Reserves	50m
	170m

- Observe from this illustration that the new capital structure remains absolutely unchanged except for narration on number of shares outstanding.

FACTORS AFFECTING DIVIDEND POLICY

- AVAILABLE INVESTMENT OPPORTUNITIES & FINANCING OPTIONS
- CLIENTELLE CONSIDERATION : Managers should consider shareholder preferences for immediate cash or capital gains. Tax status of shareholders may be important consideration
- LEGAL CONSIDERATION: Governments may enact laws or regulations restricting dividend payments to-
  - Ensure growth targets of the economy are met
  - Limit dividend repatriation
  - Strengthen capital base of critical sectors of the economy

FACTORS AFFECTING DIVIDEND POLICY

- BOND INDENTURE PROVISIONS: Managers should consider covenants with creditors to avoid unpleasant situations
- AGENCY PROBLEMS & MONITORING COSTS: Existence of information asymmetry between managers and external shareholders creates opportunity for expropriation of shareholder's wealth.
  - Dividend may be used as a monitoring mechanism to reduce agency conflict
- SIGNALLING EFFECT: Managers can use dividend as information bundle to communicate private information to critical publics of the firm and create the best possible announcement effect. They must however consider-
  - Whether dividend is leading indicator of future earning
  - What investors decode from reduction or increases in dividends

FACTORS AFFECTING DIVIDEND POLICY

- EARNING:
  - Past earning
  - Current earning
  - Future earning
- SUSTAINABILITY: Managers always try to adopt policies that stand the test of time. Hence such stable payout forms as-
  - Constant dividend amount
  - Constant payout ratio
  - Steadily incremental payout ratio

DIVIDEND POLICY MODELS

The key questions in modeling dividend policy decisions include:

- Do shareholders prefer to be rewarded by way of cash dividend or capital gain
- Does dividend policy matter? Does it have any impact on value of the firm or wealth of shareholders?

DIVIDEND POLICY: SCHOOLS OF THOUGHT

- TRADITIONAL VIEWS OF DIVIDEND RELEVANCE
  - Traditional Right
  - Radical Right
- DIVIDEND IRRELEVANCE
  - Residual theory
  - Miller & Modigliani Theorem

TRADITIONAL RIGHT VIEWS

- Dividend Policy matters, and infact, the higher the dividend payout ratio, the higher the market value of the firm
- This implies that shareholders are prepared to pay higher premium for higher payout ratio
- This is based on the legendary bird-in-hand argument since cash dividend is assumed to be less risky than potential capital gain

TRADITIONAL RIGHT MODELS

- Equity Present Value model

$$P_0 = \sum_{t=1}^{\infty} \left(\frac{D_t}{(1+ k_e)^t} \right) = \frac{D_1}{(1+ k_e)^1} + \frac{D_2}{(1+ k_e)^2} \text{------} \frac{D_\infty}{(1+ k_e)^\infty} \quad \text{(eqn. 2)}$$

Where D_t is dividend cashflow series to infinity

k_e is cost of equity or discount rate

P_0 is current equity value

- From the model, increase in dividend would result to increase in current equity value or shareholders wealth.

RADICAL LEFT VIEWS

- Dividend policy matters, in the manner that the lower the dividend payout or higher earning retention, the higher the market value of the firm.
- This implies that shareholders are prepared to pay higher premium for firms that retain more of their earning in the business
- This is typically true for growth firms where rate of return is higher than cost of capital
- It is supported within the tax state preference framework where key shareholders would prefer to be rewarded by way of capital gain

RADICAL LEFT MODELS

- Gordon's constant growth model which is an adaptation of the present value of equity model

$$P_0 = \frac{D_1}{k_e - g} = \frac{D_0(1+g)}{k_e - g} = \frac{D_0(1+kr)}{k_e - kr} \qquad \text{(eqn. 3)}$$

Where g is growth rate which may be derived by rate of return (k) multiplied by retention rate (r).

- From the model, any increase in retention or reduction in dividend payout would lead to increase in current market value (P_0)
- A similar result obtains with a more realistic supernormal growth model

$$P_0 = D_1\{1 - \left(\frac{1+g_1}{1+k_e}\right)\text{^n}\}/(k_e - g_1) + \frac{D_1(1+k_e)^{n-1}(1+g_2)}{(k_e - g_2)(1+k_e)^n} \qquad \text{(eqn. 4)}$$

MILLER & MODIGLIANI IRRELEVANCE THEOREM (1961)

- The core of M&M Thesis is that value of the firm is independent of how a firm decides to distribute its net earning between retained earning for new investments and cash dividend (Proposition 3). This implies that:
  - Dividend policy does not matter
  - Dividend policy is independent of investment and financing decision decisions
  - Two firms in the same risk class having the same operations would have identical value.

M & M ASSUMPTIONS

- Information is freely and cost-lesslly available
- There are no taxes and transaction costs
- There is certainty as to investment decisions and resulting earnings
- Investors can borrow and lend at same rate.
- Investors are rational and indifferent between capital gains and equivalent cash dividends
- Excess cash can only be invested as zero NPV projects

If these assumptions hold , it implies that in equilibrium ,

- Rate of return will be equal to cost of capital.
- Firms may decide to pay all earning as dividend without affecting ability of the firm to take on new projects, or, retain all the earning without compromising wealth of investors. This further implies that:
  - Investors can use borrowing and lending to, in place of retention and cash dividend, sustain their wealth or value

MILLER & MODIGLIANI THEOREM

- Using the above assumptions, M&M posit that two firms in the same risk class, having identical operational outlook in every parameter except for current dividend should have the same value.
- By applying the recursive single period valuation formula:

$$P_0 = \frac{D_1 + P_1}{1 + R_1} \qquad \text{(eqn. 5)}$$

Multiply both sides by n_0

$$n_0 P_0 = n_0 \frac{D_1 + P_1}{1 + R_1}$$ i.e. total market value

Given $n_0 = n_1$ *i.e. no new shares issued*

$$n_0 P_0 = \frac{n_1 D_1 + n_1 P_1}{1 + R_1} \qquad \text{(eqn. 6)}$$

Where, $n =$ *number of shares*

$P =$ *market price per share*

$D =$ *dividend per share*

$R =$ *market required rate of return*

MILLER & MODIGLIANI THEOREM

- But the firm may still decide to pay dividend(D) and also make new investment (I)
  - These can only be financed by:
    - Net operating cashflow (NOC)
    - Issue of new shares (m)
    - Hence $NOC_1 + m_1 P_1 = I_1 + n_1 D_1$ (eqn. 7)
      - » $m_1 P_1 = -NOC_1 + I_1 + n_1 D_1$ (eqn. 8)
    - If new shares are issued, $n_1 = n_0 + m_1$ (eqn. 9)
- Therefore: $n_0 P_0 = \frac{n_1 D_1 + n_1 P_1 - m_1 P_1}{1+R_1}$ (eqn. 10)
  - $n_0 P_0 = \frac{n_1 D_1 + n_1 P_1 - (-NOC_1 + I_1 + n_1 D_1)}{1+R_1}$
  - $n_0 P_0 = \frac{n_1 D_1 + n_1 P_1 + NOC_1 - I_1 - n_1 D_1)}{1+R_1}$
  - $n_0 P_0 = \frac{n_1 P_1 + NOC_1 - I_1}{1+R_1}$ (eqn. 11)
  - Dividend has been eliminated from the equation.
- Accordingly market value of a firm does not depend on dividend but on cashflows arising from its operations.

ILLUSTRATION 3

- Durella Ltd's financial statement shows share capital of N100,000 nominal value of N1.00 per share. Net operating income is estimated to be N300,000. Meanwhile the company has an investment opportunity of N500,000 for which new shares can be issued at par to finance it. Given the assumptions of Miller & Modigliani and company's cost of capital of 10%, prove that dividend policy is irrelevant to the firm's financial management decisions.

ILLUSTRATION

Durella Ltd

P_1 = N5

NOI_1 = N300,000

n_0 = 100,000 shares

$n_1 = n_0 + m_1$ = 200,000 shares

R = 10%

I_1 = N500,000

- If no dividend is paid out of NOI:

$$n_0P_0 = \frac{NOI_1 + n_1P_1}{1+R_1} \qquad \text{(eqn. 12)}$$

$$100{,}000P_0 = \frac{300{,}000 + 100{,}000(5)}{1.1}$$

P_0 = N7.27

- If NOI is paid out as dividend, new issues are made to fund replacement investment:

$$n_0P_0 = \frac{n_1P_1 + NOC_1 - I_1}{1+R_1} \qquad \text{(eqn. 13)}$$

$$100{,}000P_0 = \frac{200{,}000(5) + 300{,}000 - 500{,}000}{1.1}$$

P_0 = N7.27

- Hence dividend policy does not matter

HYBRID MODEL

- WALTER'S MODEL proposes an intuition that relationship between dividend policy and value is conditional on relationship between rate of return (K) and cost of capital (k_e).
- $$P_0 = \frac{D_1 + \frac{k}{k_e}(E_1 - D_1)}{k_e}$$ (eqn. 14)

WALTER's MODEL

- SCENARIO I: Where K > k_e (growing firms)
  - It pays the firm to retain more cash in the firm to take on new investments. Hence less dividend payout will lead to higher value
  - Investors would prefer capital gain potential to cash dividend
  - Infact 100% retention will maximize shareholders' wealth
  - This scenario is consistent with Radical Left model and dividend policy matters

WALTER's MODEL

- SCENARIO ll: Where K < k_e (declining firms)
  - It pays the firm to pay more dividend to shareholders
  - Infact 100% dividend payout or 0% retention will maximize wealth for shareholders
  - Shareholders would prefer cash to potential for capital gain
  - This scenario is consistent with Traditional Right model and dividend policy matters

WALTER's MODEL

- SCENARIO lll: Where K = k_e (Normal firms)
  - Here the firm would be indifferent between dividend and capital gain potential
  - Dividend policy does not matter and would not affect value of the firm
  - This is consistent with Miller & Modigliani Thesis.

ILLUSTRATION 4

- Unilag Dividend Ltd has over the past 5 years recorded average annual earning of N500m on the back of total investment worth N2.5b. The company maintains constant dividend payout ratio of 60%. If the firm's cost of capital for firms in the same risk class is 15%:
  - a) Determine the current market price of the firm.
  - b) Consider a sister company, Unilag Growth Ltd, with identical operating statistics but for payout rate of 40%. Would the market price be different from (a) above?
  - c) Use results in (a) and (b) above to discuss relevance of dividend policy decisions

ILLUSTRATION
(Unilag Dividend Ltd)

- $P_0 = \frac{D_0(1+g)}{k_e - g}$
- K = $\frac{NOE}{Investment} = \frac{N500m}{N2{,}500m} = 20\%$
- r = 100% - 60% = 40%
- R = 15%
- D_0 = NOE(1-r) = N500m(60%) = N300m
- g = kr = 0.2(0.4) = 0.08
- $P_0 = \frac{300m(1.08)}{0.15-0.08} = \frac{N324m}{0.07}$ = **N4,629m**

ILLUSTRATION
(Unilag Growth Ltd)

- $P_0 = \frac{D_0(1+g)}{k_e - g}$
- K = 20%
- r = 100% - 40% = 60%
- R = 15%
- D_0 = N500m(40%) = N200m
- g = kr = 0.2(0.6) = 0.12
- $P_0 = \frac{N200m(1.12)}{0.15-0.12} = \frac{N224m}{0.03} = \boldsymbol{N7,467m}$
- Clearly the illustration shows that a higher retention rate gives rise to higher market value.

- The results above support the dividend relevance views of the radical left. In consonance with this school of thought, a reduction in dividend payout rate from 60% to 40% resulted in increase of market value from N4,629m to N7,467m.

ASSIGNMENT 5

- A) Given that all operational statistics remain the same and rate of return on investment was 10%, would solutions to questions a, b and c in illustration above be different?
  - Unilag Dividend Ltd N1,418m
  - Unilag Growth Ltd N1,178m

ASSIGNMENT 6

- Given that average annual earning for Unilag Dividend Ltd and Unilag Growth Ltd reduced to N250m, while other financial data remain the same, use Walter's Model to compare with results of:
  - a) Unilag Dividend Ltd at N500m annual earning
  - b) Unilag Growth Ltd at N500m annual earning

SOLUTION

- Unilag Dividend Ltd (60% Payout)
  - At N500m: P_0 = N4,080m
  - At N250m: P_0 = N1,502m
- Unilag Growth Ltd (40% Payout)
  - At N500m: P_0 = N4,480m
  - At N250m: P_0 = N1,413m

- At higher earning rate, it pays to payout less dividend. At lower earning rate, it pays to pay more dividend to shareholders.